



Stochastic dual-scale dynamics of visual inference

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Declaration

I, *Jude Luca Metcalf*, declare that this thesis is submitted in partial fulfilment of the requirements for the conferral of the degree *Bachelor of Science (Honours)*, from the University of Sydney, is wholly my own work unless otherwise referenced or acknowledged. This document has not been submitted for qualifications at any other academic institution.

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Abstract

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Introduction

The brain is among the most widely studied subjects of research, and yet many neural behaviours remain elusive to scientists. In recent years, methods adapted from statistical physics, network science, and thermodynamics have been found to explain and provide understanding for computation in the brain. A large part of the brain is devoted to vision, the visual cortex containing between 4-6 billion of the 30 billion neurons of the neocortex, up to 5 times the size of other primates [1, 2]. Vision is thus an important neural process, and provides one of the first gates of attention to the human brain.

Many approaches to understanding patterns of human gaze have been described. Importantly, there has been success in training large neural networks to predict gaze trajectories [3–8] and saliency maps from an image for human observers [9–14]. Such approaches give often quite accurate results, but they lack the rich understanding of a full model of the network of the brain itself. Theories of biological attention and cognition, such as those of the “Bayesian brain”, built on active sensing and inference provide a more motivated framework to understand the brain as maintaining a rational causal mental worldview [14–17]. This Bayesian model of the brain has been a fruitful direction to understand the most basic level of neural cognition as sampling through population activity patterns [18–21]. Although it has been a debated framework (see [22]), the Bayesian brain has provided a strong window into biological cognition. Geoffrey Hinton and colleagues invented the Helmholtz machine, performing explicit Bayesian inference on perceptive inputs, and updating predictions on the external world, minimising the free energy as a loss function of a neural network. Such a formulation inspired pioneering work on neural coding [20, 23], Karl Friston’s highly influential Free Energy Principle in theoretical neuroscience [24, 25], and the resulting empirical models and well-verified predictions about the human sensory-motor system [26–28].

Chapter 1: Survey of the Literature

This thesis approaches the problems of computation in the brain and in machine learning through understanding the rich spatiotemporal dynamics of visual searching. We use the stochastic FNS model, studying the properties of anomalous diffusion and complex multiscale spatiotemporal dynamics it yields, with good theoretical background in the neural circuit model it was developed for. We aim to understand these behaviours as an efficient method of gathering information. In Section 1.1 we introduce the modelling framework of stochastic calculus and Lévy motion which informs the dynamics of the FNS sampler. Section 1.2 then provides the link between stochastic dynamics and computation sampling, both in biological and abstract contexts. Section 1.3 provides the background for much of the behavioural data we present and model in this thesis, and the previous methods used to understand and model the complex dynamics found in gaze trajectories especially.

1.1 Stochastic Modelling

Stochastic calculus has been fundamental in understanding how high dimensional complex systems reduce to a small number of variables of interest. When studying complex systems in physics, economics, geology, climate science, machine learning, and neuroscience, it is often the most natural, tractable, and accurate modelling choice when understanding a system of interest. The important concept here is that of the introduction of some central limit theorem, where the sum of very many small interactions which are understood microscopically can produce certain well-defined stochastic patterns on macroscopic scales of interest. The ubiquitous stochastic process is a Wiener process W_t , equivalently called Brownian Motion. A Wiener process is defined by the stationary and independent Gaussian increments across all non-overlapping time intervals. The Wiener process is a very simple and analytically elegant model of the coarse-grained effect of very many small finite-variance interacting influences. A Wiener process is defined only by its timescale, with a variance over trajectories after time t of t itself. Wiener processes yield time series which are almost surely continuous but nowhere differentiable. The Wiener process is understood as a model of the long-time dynamics of a variable effected by very many small uncorrelated finite variance influences [29]. Classical examples of processes modelled by the Wiener process are: the movement of a macroscopic particle in a non-turbulent fluid [29], the gravitational influence of an orbiting thermal collection of stars on supermassive black holes [30, 31], and (arguably) the movement of low-volatility stock prices [32]. There are well studied mechanisms through which complex real-world stochastic processes diverge from the Wiener process. Of interest to us is the case where the microscopic effects become correlated with heavy-tailed statistics. Correlated steps, long-time memory effects, and microscopic dynamics can result in a stochastic process which does not limit to the Wiener process [33, 34]. Of interest to us is the fractional and superdiffusive Lévy flight, which is a result of spatial non-locality in an interacting network. For instance, previous studies have demonstrated a neural circuit network at criticality resulting in diverging correlation length of activity, and the centre of mass of neural firing is subject to the large jumps. Such large jumps in a stochastic time series are modelled according to Lévy stable motion, where the increments are drawn independently from a heavy-tailed distribution. This section describes the analytics and behaviour of such Lévy processes for our current study.

1.1.1 Anomalous Superdiffusion

A key motivation for the development for stochastic modelling was when studying processes of diffusion. In 1885, the German physiologist Adolf Fick was the first to describe the universality of normal diffusion [35]. Fick records the diffusion equation, which is identical to the heat equation applied to a probability distribution, as

$$\frac{\partial p(x, t)}{\partial t} = D \frac{\partial^2 p(x, t)}{\partial x^2} \quad (1.1)$$

where D is a scaling constant. Fick shows that this gives a mean squared displacement (MSD) $\langle x(t)^2 \rangle = 2Dt$. This MSD linear in time defines normal diffusion. Such normal diffusion is found in many simple random processes, such as the paradigm case of the Brownian motion of dust in a fluid studied by Albert Einstein, the motion of nutrients through cell membranes, and the movement of a random walk. The modelling work of Albert Einstein and the British statistician Karl Pearson both independently have found that modelling such a process as a continuous time random walk (CTRW) will yield this normal diffusion, given the distributions of step length and time between steps are both finite variance. Increasingly, physical systems have been studied which violate these properties. Such a case is known as anomalous diffusion, giving a MSD which is nonlinear in time:

$$\langle x(t)^2 \rangle \propto t^\beta, \quad \beta \neq 1 \quad (1.2)$$

Such anomalous diffusion yields two cases, subdiffusion $0 < \beta < 1$ where diffusion scales slower than normal, and superdiffusion $1 < \beta < 2$ where diffusion scales faster than normal. The case of $\beta = 2$ is known as ballistic diffusion, as it is found in the deterministic case of unchanging velocity $x = vt$. [THE CASE OF SUBDIFFUSION IN FLUID FLOW, ETC]. Superdiffusion has been studied in the contexts of complex biological systems, especially in the context of searching for scarce resources. The hypothesis that a superdiffusive CTRW (known as a Lévy walk) is an optimal strategy when searching for sparse and randomly distributed resources has been well studied, debated, and verified in certain contexts as the Lévy foraging hypothesis [forag].

Simulating trajectories from stochastic processes such as Wiener processes and Lévy flights are computationally direct, it is often useful to understand the dynamics of the distribution of trajectories after some time t . The dynamics of the distribution of trajectories driven by a stochastic differential equation is defined through the Fokker-Planck Equation (FPE), also known as the Kolmogorov Forward Equation. The paradigmatic FPE is given for the Langevin Equation:

$$dX_t = \mu(X_t, t)dt + \sigma(X_t, t)dW_t \quad (1.3)$$

giving the partial differential equation for the distribution $p(x, t)$ of X_t :

$$\frac{\partial p(x, t)}{\partial t} = -\frac{\partial}{\partial x}[\mu(x, t)p(x, t)] + \frac{1}{2} \frac{\partial^2}{\partial x^2}[\sigma^2(x, t)p(x, t)] \quad (1.4)$$

In the case where $\mu(x, t) = 0$ and $\sigma(x, t) = \sigma$, the Langevin FPE simplifies into the standard heat equation, with diffusion of the probability mass with a rate defined by the volatility parameter σ :

$$\frac{\partial p(x, t)}{\partial t} = \frac{\sigma^2}{2} \nabla^2 p(x, t) \quad (1.5)$$

The heat equation is very well studied, and given an initial condition concentrated at the point $(x, t) = (x_0, 0)$, the distribution at any time $t > 0$ is exactly a Gaussian defined by:

$$p(x, t) = \frac{1}{(2\pi\sigma^2 t)^{1/2}} \exp\left(-\frac{|x|^2}{2\sigma^2 t}\right) \quad (1.6)$$

We see from this that the variance of such a distribution will scale as $\mathbf{Var}_x[p(x, t)] \propto t$

1.1.2 Lévy Stable Motion

The literature is often unclear when distinguishing stochastic processes of a heavy-tailed nature. However, for our purposes, we analyse what is often called the *Lévy flight* L_T^α , a Stochastic process defined through the increments being drawn from the heavy tailed symmetric α stable distribution (a $\mathcal{S}\alpha\mathcal{S}$ distribution). $\mathcal{S}\alpha\mathcal{S}$ distributions are a family of limiting stable distribution under summation of i.i.d infinite-variance variables, as a result of the generalised central limit theorem. The α value defining the particular $\mathcal{S}\alpha\mathcal{S}$ distribution defines the asymptotic tail of the probability density function, with $p(x) \propto |x|^{-1-\alpha}$. Thus, the Lévy flight L_t^α after time t is defined as follows, to outline the fractional scaling properties and alpha-stable statistics

$$L_t \stackrel{d}{=} t^{1/\alpha} L_1, \quad L_1 \sim \mathcal{S}\alpha\mathcal{S} \quad (1.7)$$

where the parameter α defines the heaviness of the tails of resulting statistics. In the limit $\alpha \rightarrow 2$, the Lévy flight reduces to a scaling of Brownian Motion $L_t^{\alpha \rightarrow 2} = \frac{1}{\sqrt{2}} W_t$. If $1 < \alpha < 2$ the variance of L_t^α diverges, but L_t^α has 0 mean, whereas if $0 < \alpha \leq 1$, then both the variance and mean of L_t^α diverge. Many of the simple analytic properties of the Wiener process are no longer true of a Lévy flight. Importantly, trajectories of a Lévy flight are almost surely no longer continuous. Instead, such trajectories are called *càdlàg* (continue à droite, limite à gauche) with respect to time, which describes functions which are right continuous with well-defined left limits [36]. Thus, we say that since all such trajectories produced from a Lévy flight are càdlàg, the Lévy flight itself is called a càdlàg process. Càdlàg processes have been widely studied in the context of mathematics, economics, and physics, and their properties are well understood [37, 38]. It has been shown that Lévy flights are semimartingales, and are well interpreted in integration [36]. Importantly, it follows from the theorem of the Lévy-Itô decomposition that a Lévy flight can be decomposed as the sum of two processes [39]:

1. A compound Poisson process of large jumps with $|\Delta x| > 1$. Such jumps are exponentially distributed in time, and there are almost surely a finite number of such jumps in any finite time period.
2. A square-integrable martingale jump process with $|\Delta x| < 1$. This process is thus finite-variance, and when coarse-grained over time, approaches a Wiener process.

This division is useful in analysing the process in the context of stochastic nonlocal global exploration, which is modelled by the compound Poisson process of large jumps $|\Delta x| > 1$, and the local exploitation dynamics modelled by the square-integrable martingale of small jumps $|\Delta x| < 1$.

1.2 Abstract Computation

Probabilistic sampling is a powerful and flexible paradigm of general computation. Sampling methods represent and evaluate probability distributions through the production of

stochastic samples. Uncertain, high-dimensional, and ambiguous inference problems require efficient and accurate characterisations of target distributions. Sampling methods draw from a target distribution such that the sampled ensemble distribution reflects the target itself. This paradigm has become widely utilised in machine learning and statistics, where generative models learn patterns from large multivariate distributions inferred from observational data [40, 41]. Recently, this framework has gained significant traction as a model for biological computation. The position that the brain itself acts as a Bayesian generative model has become that interprets through continuous sampling mechanisms has become well supported.

1.2.1 Active Searching and Inference

The dominant framework in modern computational neuroscience is that of the Bayesian brain. Karl Friston, the most cited neuroscientist [42], developed this framework, and shifted the view of the perceptual loop of the brain. Friston’s theory of active inference described perception through inferring and generating top-down sensory predictions of the world, and carrying back the bottom-up errors from that perception. This development marked a strong shift in the literature, “marking a watershed between 20th-century thinking about the brain as a glorious stimulus–response link and more constructivist 21st century perspectives that emphasized an active sampling of the sensory world” ([43] pg. 1019).

1.2.2 Neural Sampling

The neural sampling hypothesis posits that the variability of a cortical neural response is an important functional feature of the biological computational sampling. Previously regarded as noise, the neural sampling framework interprets instantaneous neural activity patterns as stochastic samples drawn from a posterior distribution over possible sensory interpretations [18, 19]. As such, the location and frequency of firing patterns across cortical neurons are interpreted as stochastically sampling an informed target posterior, acting as the basis for some kind of computation. Consequently, perceptual uncertainty is directly encoded by the variability of cortical responses across time and across multiple trials, with higher uncertainty reflected in wider response distributions [18, 19, 40]. Neural sampling thus provides an interpretation for the way the brain navigates uncertainty at the fundamental scale of computation. Various biological mechanisms have been proposed to explain how neural circuits generate samples.

Traditional models often assume the existence of explicit stochastic noise sources, however recent models demonstrate that strongly recurrent, balanced inhibitory-excitatory deterministic neural networks can generate the high-dimensional variability which stochastic models marginalise over through chaotic dynamics. These emergent chaotic dynamics facilitate Bayesian learning, as for instance described by Terada and Toyozumi through training a recurrent network to sample static features and dynamic trajectories, generalising learned priors to novel, uncertain environments through biologically plausible learning [44]. Furthermore, energy-based models (EBMs) have been developed which suggest that fast-spiking interneurons can estimate the local partition function to enable fast, localised sampling dynamics [45]. The sampling hypothesis has thus given a convincing theory of biological neural computation when contrasted with the trial-averaged perspective of some averaged response combined with noise linked to neural variability [18]. However, traditional Gaussian or Poisson noise models representing these stochastic network dynamics inherently correspond to continuous Brownian motion and struggle to capture

the full dynamics of a complex interconnected inhibitory-excitatory biological neuronal network [46–48].

Fractional Neural Sampling

In reality, neural activity exhibits structure across multiple spatial and temporal scales and exhibit heavy-tailed, non-Gaussian statistics. This observation motivates the design and theory of the theory of Fractional Neural Sampling (FNS). FNS describes the complex spatiotemporal dynamics of spiking neural circuits, where localised activity patterns propagate not through ordinary diffusion, but via superdiffusive Lévy flights. Such dynamics are called fractional, as they are the result of nonlocal interactions requiring fractional calculus to describe [49]. These which can be mathematically characterised by stochastic differential equations with fractional order derivatives. The critical advantage of FNS over standard continuous random walks is its ability to efficiently sample complex, multimodal probability distributions with far-apart modes—a major challenge when processing natural environments replete with multiple distinct salient features. While Brownian-motion-based samplers typically remain trapped in local modes due to high potential energy barriers, the heavy-tailed, large jumps inherent in FNS enable the neural sampler to adaptively and freely switch between multiple modes [50–53]. This dynamical switching provides a mechanistic explanation for the bimodality of estimate distributions observed in human perceptual switching and visual perception inference.

1.2.3 Markov Chain Monte Carlo Methods

Markov Chain Monte Carlo (MCMC) methods constitute the algorithmic foundation for generating samples from target distributions in Bayesian inference and machine learning. MCMC operates by constructing a Markov chain whose equilibrium distribution is the desired posterior distribution. In the context of neural computation, biological circuits are often modelled as implementing continuous-time variants of MCMC, most notably Langevin sampling and Hamiltonian Monte Carlo (HMC). Langevin dynamics utilise local gradient information from the probability landscape, combined with Gaussian white noise, to explore the state space over time. Despite their ubiquity, traditional MCMC methods encounter severe theoretical and practical limitations when dealing with multimodal distributions. Standard Langevin and HMC dynamics are driven by continuous Brownian and Hamiltonian motion respectively, and the traversal of low-probability regions, equivalent to dynamical energy barriers, becomes exponentially slow. Consequently, the mean exit time for the sampler to leave one mode and discover another grows exponentially with the modal separation distance. To accelerate this sluggish exploration, algorithms often introduce auxiliary momentum variables. However, even with momentum-based acceleration, traversing deep energy valleys remains highly inefficient [50, 51]. The fractional dynamics introduced in FNS-style sampling are one area of research into how fast mixing of samples in complex distributions can be achieved without the need for non-trivial parameter tuning. For instance, algorithms based on Hamiltonian Monte Carlo sampling and second-order Langevin sampling very similar to FNS dynamics have been briefly developed and explored [52, 54, 55]. Under FNS dynamics, the mean exit time between modes scales linearly rather than exponentially with separation, demonstrating a profound computational advantage for efficient biological Bayesian inference [55].

1.3 Behavioural Data

1.3.1 Neural Frequency Bands

Throughout much of this thesis, links are drawn between timescales of interactions and dynamics in the FNS model and well-studied frequency bands of activity in the human brain. It is then valuable to outline the studied computational and functional interpretations of such frequency bands. Neural firing activity is divided into frequency bands named by greek letters, originally discovered and decomposed from electroencephalogram (EEG) data. Divisions are only approximate, but for this thesis these bands are: delta (0.1 - 4 Hz), theta (4 - 8 Hz), alpha (8 - 14 Hz), beta (14 - 30 Hz), and gamma (30 - 220 Hz) [56, 57]. Of particular interest to this thesis are the interpretations of theta and gamma neural activity as functionally important to attention and search tasks of computation. Theta rhythms have been shown to modulate attention during auditory tasks [58, 59], visual search tasks [60–62], and in analysing the general attentional mechanism of the brain [17, 56, 57]. Intervals between theta peaks are finding stronger interpretation in the neuroscience literature as discretely sampling the environment for integration into the internal model of the brain. Strong direct evidence for this theory include correlations between intelligence and high theta peak power in EEG data during tasks requiring focus and attention [17, 58], and the discovered sensitivity of the visual system to react and search scenes on a theta rhythm [60, 61]. Accordingly, the lower level attentive activity which is modulated by theta is regularly found to be gamma [63–67].

1.3.2 Gaze Trajectories

Healthy human gaze trajectories consist of different dynamical regimes. The most studied property of the pattern of gaze trajectories are their division into periods of fixation, interspersed by near-instantaneous jumps between fixations, these jumps being known as “saccades” [68–71]. Importantly, this behaviour of local movements interspersed with large discontinuous jumps suggest modelling eye movements through a Lévy flight. This method has a limited history in the literature, and fitting data to a simple phenomenological Lévy flight parametrised by the stable parameter α found limited use as a biomarker for ADHD [72]. Saccade metrics in general have been found to be valuable as non-invasive biomarkers for broader neural disorders, including schizophrenia, autism, ADHD, OCD, bipolar disorder, Parkinson’s disease, and multiple sclerosis [73–77]. Periods of fixation have received less scholarly attention as a biomarker of cognition. Gaze fixations exhibit three distinct dynamical patterns: near-instant microsaccades within the same point of focus, slow ocular drift about a point of focus, and high frequency ocular micro tremor [78–81]. Examinations and literature reviews of these behaviours have often not considered their computational role in active sensing and efficient search.

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